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DIFFERENTIAL ROTATION OF THE EASTERN PART OF SOUTHWEST JAPAN INFERRED FROM PALEOMAGNETISM OF CRETACEOUS AND NEOGENE ROCKS

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Abstract. Paleomagnetic directions have been determined on Late Cretaceous and Miocene rocks in the Nohi area, in the eastern part of southwest Japan, in an effort to clarify within-block deformation of the rotated part of southwest Japan. Twenty-two sites in the area are found to have a reliable primary magnetic component through progressive demagnetization tests using thermal and alternating field methods. Tilt-corrected paleomagnetic data in the Nohi area are divided into three stages based on geologic age: Late Cretaceous, early Miocene (23-15 Ma), and middle Miocene (12-10 Ma). The formation-mean direction of 10 sites obtained from Late Cretaceous Nohi rhyolite is D=17.0°, I=49.8°, and α_{95}=6.1°. Neogene rock units underlain by Nohi rhyolite show two distinct formation-mean directions: D=13.0°, I=52.7°, and α_{95}=7.2° for eight sites of early Miocene, and D=-2.5°, I=55.6°, and α_{95}=6.6° for four sites of middle Miocene. Differential rotation is proposed between the eastern part of southwest Japan including the Nohi area and the central part of southwest Japan, which is known to have rotated clockwise through about 50° around 15 Ma. The spatial distribution of contemporaneous paleomagnetic data in southwest Japan suggests that two differentially rotated parts are connected in the western neighborhood of the Nohi area. It is concluded that the Nohi area had rapidly rotated counterclockwise through 48° relative to the central part of southwest Japan between 15 and 12 Ma. Together with previous paleomagnetic data from early Miocene rocks in the Kanto area, which is situated to the east of the Nohi area, the present results suggest that differential rotation of the eastern part of southwest Japan is attributed to the formation of northward cuspate shapes of pre-Neogene terranes which constitute the framework of the Japan arc. One probable explanation of the formation of cusps is a crustal indentation caused by the collision of landmasses on the Philippine Sea plate against the Japan arc in the middle Miocene.

Introduction

The Japan arc-Japan Sea pair is one of the most intensely studied island arc-back arc basin systems in the western Pacific (Figure 1). The Japan arc is divided into southwest Japan, west of the Itoigawa-Shizuoka tectonic line (ISTL), and northeast Japan, east of the ISTL (Figure 1), on the basis of Neogene tectonics. Recent paleomagnetic studies [Otofuji et al., 1985a,b] have revealed that southwest Japan and the northern part of northeast Japan, north of the Tanakura tectonic line (TTL), had undergone clockwise and counterclockwise rotation, respectively, associated with the back arc opening of the Japan Sea. Based on a data set obtained from southwest Japan since 1982, Otofuji et al. [1985a] estimated the amount of clockwise rotation as 47°. They decided the timing of clockwise rotation of southwest Japan using best fitting curves for rotation deduced from declination versus age data; the climax of rotation is at 14.9 Ma, and its duration is 0.6 m.y. As for the amount of clockwise rotation, Otofuji and Matsuda [1987] carried out paleomagnetic studies in the San'in area, which is a rather limited area in the central part of a rotated continental sliver of southwest Japan (Figure 1), in order to obtain characteristic paleomagnetic directions for the southwest Japan block free from the effects of different block motions. They concluded that southwest Japan had been rotated through more than 40° with respect to the eastern margin of Eurasia by a fan-shaped opening of the southwestern part of the Japan Sea. On the contrary, some paleomagnetic data in the eastern part of southwest Japan suggest a much smaller rotation angle. The mean paleomagnetic declination of Oligocene-early Miocene volcanic rocks in the Noto Peninsula (Figure 2) shows no significant deflection [Hirooka et al., 1986]. Itoh [1986] carried out a paleomagnetic study on the Miocene volcanic and sedimentary rocks collected from the Higashitonami hilly area near Yatsuo (Figure 2) and concluded that the area had rotated clockwise about 13° between 15 and 12 Ma. The smaller rotation angle suggests deformation within southwest Japan since middle Miocene. Though the deformation mode of the Japan arc was discussed on the basis of geological evidence [Faure and Lalevee, 1987], the deformation feature in the eastern part of southwest Japan and the western part of northeast Japan has not yet been clarified because of its complicated structure. Because southwest Japan basement partly consists of Paleozoic to Mesozoic metamorphic complex and granitic rocks, it can be regarded as a microcontinent. Therefore descriptions of the intradeformation of southwest Japan will not only fix the tectonic history of Japan but also shed some light on the physical properties of the continental lithosphere, e.g., ductility. In this study, extensive paleomagnetic data in the eastern part of southwest Japan are presented. The data describe differential rotation between the eastern and the central part of southwest Japan which occurred contemporaneously with the opening of the Japan Sea. Possible arc-arc collision events which raised the crustal indentation and the differential rotation in the Japan arc are discussed from the viewpoint of the shape of the indentation and timing of the collision.

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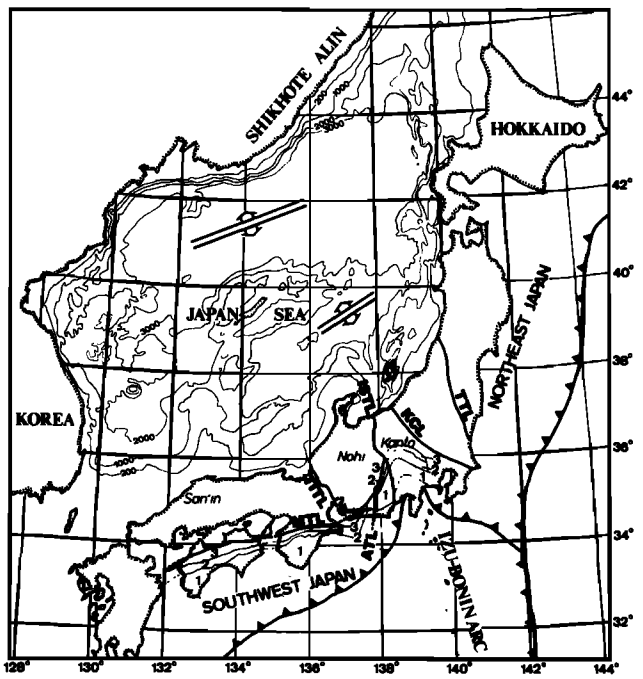


Fig. 1. Index map of geotectonic divisions in Japan with the distribution of pre-Neogene (1) Shimanto, (2) Chichibu, and (3) Sambagawa terranes. Abbreviations are TTL, Tanakura tectonic line; KCL, Kashiwazaki-Choshi line; ISTL, Itoigawa-Shizuoka tectonic line; ATL, Akaishi tectonic line; TITL, Tsurugawan-Isewan tectonic line; and MTL, Median tectonic line. Base map shows contours and possible position of remnant spreading centers in the Japan Sea. The present trench system around the Japanese Islands is also shown. Compiled from Yamada et al. [1982], Research Group for Active Faults [1980], American Association of Petroleum Geologists [1981], and Takeuchi [1986].

Geological Setting and Sampling

The present paleomagnetic investigation is focused on Late Cretaceous volcanic rocks that were collectively named the Nohi rhyolite by Kawada et al. [1961], and the Neogene strata around Yatsuo (Figure 2). The Nohi rhyolite consists mainly of rhyolitic welded tuffs, which were autochthonously formed on Paleozoic and Mesozoic terranes. Because the Nohi rhyolite has suffered no serious deformation or fracture, older terranes covered by this widespread rock body can be regarded as a stable and rigid block since Late Cretaceous, which is called the Nohi area in this study (Figures 1 and 2). The Neogene strata sampled in this study are situated at the northern rim of the Nohi area and deposited unconformably on the Nohi rhyolite. The recent paleomagnetic data, together with those of Neogene rocks in the Nohi area published previously [Itoh, 1986], describe differential rotation of the eastern part of southwest Japan since the Cretaceous.

Late Cretaceous Rock Units (Nohi Rhyolite)

Stratigraphically, Yamada et al. [1971] divided the Nohi rhyolite into five units using the lacustrine sediments as time markers. They interpreted those units to represent five stages of volcanic activities, which erupted ash flows. The lower part of the Nohi rhyolite has been dated by Rb-Sr and fission track methods as shown in Figure 2. The Rb-Sr age of the welded tuff in the lowest part (stage I) was obtained as 81.2 Ma [Shirahase, 1984]. The welded tuffs of stage II have been dated by the Rb-Sr method [Okamoto et al., 1975; Shirahase, 1984] and fission track method [Yamada et al., 1985]. Their ages range from 69.6 Ma to 72.9 Ma. Rb-Sr [Shirahase, 1984] and fission track [Yamada et al., 1985] ages of the welded tuffs in stage III are 71.2 Ma and 64.5 Ma, respectively. These ages indicate that the main part of the Nohi rhyolite was formed in the Late Cretaceous. Plant remains found from stage I rock units were also assigned to the Late Cretaceous (Campanian-Maastrichtian) [Kawada et al., 1976].

The present paleomagnetic sampling was carried out at 15 sites belonging to stage II or III (Figure 2). Samples were collected only from the following rocks: (1) tilting of rocks was clearly identified through the eutaxitic structure of welded tuff and (2) they had not been thermally and/or chemically altered by associated intrusive rocks. Each site comprised eight to 11 hand samples which were independently oriented using a magnetic compass.

Neogene Rock Units

Lower Miocene volcanics (so-called Green Tuff) and Miocene to Pleistocene marine sedimentary rocks are distributed in the northern part of the Nohi area. These rocks generally dip 10° to 40° to the north with neither serious tectonic disturbance nor slumping structure. In this study, paleomagnetic samples were taken from the hills around Yatsuo in which paleomagnetic studies had been carried out [Itoh, 1986]. The Neogene System is lithologically divided into the following eight formations: Tori, Nirehara, Iwaine, Iozen, Kurosedani, Higashibessho, Tenguyama, and Otogawa, in ascending order [Sudo, 1979; Hayakawa, 1986]. The lithofacies of these units are shown in Figure 2.

Tsuchi [1981] assigned the Kurosedani Formation to the N8 planktonic foraminiferal zone (15.5-16.5 Ma) of Blow [1969]. Based on the diatom biostratigraphy [Ito, 1986], the Higashibessho Formation was assigned to the Denticulopsis lauta zone (14.85-15.70 Ma) defined by Koizumi [1985]. From the middle part of the Otogawa Formation, Ito [1986] reported a diatom assemblage which suggested a correlation with the Thalassionema schraderi zone (7.30-9.20 Ma) of Koizumi [1985]. Many radiometric ages have been reported from the Neogene System (Figure 2). Two welded tuff samples of the Tori Formation have been dated as 22.2 Ma and 23.7 Ma using the fission track method [Ganzawa, 1983]. A K-Ar age was determined to be 16.4 Ma for the andesite lava from the Iwaine Formation [Shibata, 1973]. A fission track age was determined to be 12.3 Ma for a pumice tuff from the lower part of the Tenguyama Formation [Hayakawa, 1983]. A tuff layer from the middle part of the Otogawa Formation was dated as 9.5 Ma using the fission track method [Sumii, 1986]. These radiometric ages are concordant with biostratigraphic age estimation.

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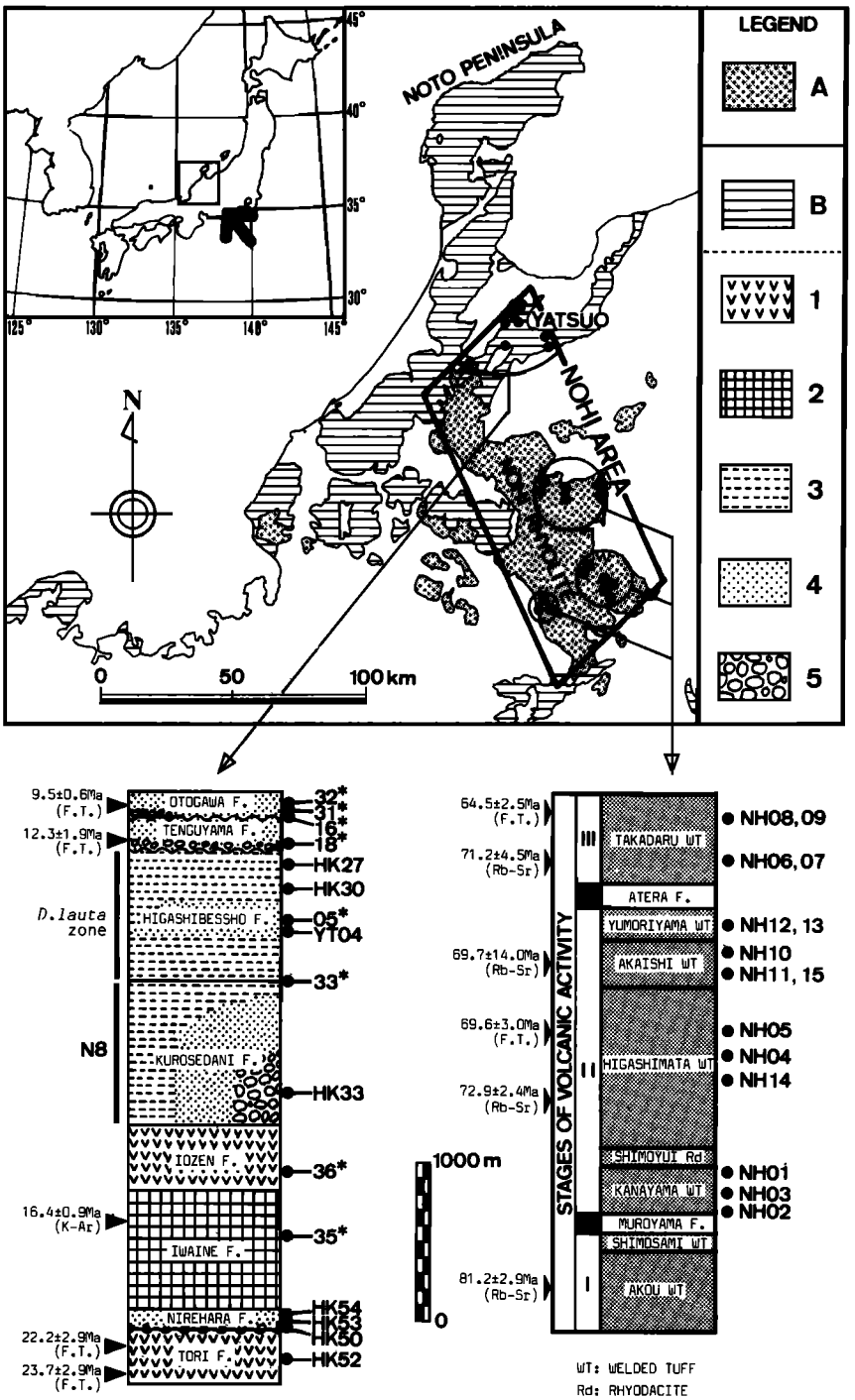


Fig. 2. Distribution of (A) Cretaceous volcanic rocks and (B) Neogene System in the eastern part of southwest Japan simplified from Yamada et al. [1982], and stratigraphic successions of the studied Nohi area after Yamada et al. [1971, 1985], Hayakawa [1986], and Sudo [1979]. Solid circles in the map indicate paleomagnetic sampling sites. Radiometric ages [Shibata, 1973; Okamoto et al., 1975; Hayakawa, 1983; Ganzawa, 1983; Shirahase, 1984; Yamada et al., 1985; Sumii, 1986], biostratigraphic ages [Tsuchi, 1981; Ito, 1986], and the horizons of sampling sites are attached to the columnar sections. Numbers denoted by asterisks show the horizons of paleomagnetic data from Itoh [1986]. 1, rhyolitic volcanic rocks; 2, andesitic volcanic rocks; 3, mudstone; 4, sandstone; and 5, conglomerate.

Geologic maps of 1:25,000 scale [Hayakawa, 1985], 1:50,000 scale [Sakamoto and Nozawa, 1960], and 1:62,500 scale [Ganzawa, 1983] provide regional structure and stratigraphy of the sampling sites. Samples were collected at eight sites as shown in Figure 2. Each site was comprised of four to 10 oriented hand samples. In each site, strike and dip of the strata were

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measured for tilt correction. Based on field observation and microscopic study, it has been certified that all sampling sites of this study did not severely alter.

Sampling Criteria for Reliable Tilt Correction

In the case of volcanic rocks, dispersion of paleomagnetic directions may be induced by difficulty in distinguishing tectonic tilt from primary dip, as suggested by Hildreth and Mahood [1985]. To perform a reliable tilt correction, it must be confirmed that primary dip is in a negligible amount in the sampling points. In this study, sampled units of volcanic rocks satisfied the following criteria, indicating negligible primary dip: (1) observed dips of volcanic rocks are consistent with the dips of sedimentary rocks which intercalate or conformably overlie/underlie the sheet of volcanic rocks, and (2) sampling sites are not stratigraphically situated at the basal units of volcanic flow deposits, which often have steep (>10°) primary dip reflecting buried topography [Hildreth and Mahood, 1985].

Experimental Procedure and Results

Two or three cylindrical specimens of 25 mm in diameter and 22 mm in length were cut from each hand sample. Remanences were mainly measured by the ScT C-112 cryogenic magnetometer. A Schonstedt SSM-1A spinner magnetometer was used for measurement of volcanic samples which had relatively high magnetic intensity.

Two specimens were selected from each site, and their magnetic stability was tested by means of progressive demagnetization with thermal and alternating field methods. Alternating field demagnetization (AFD) was carried out stepwise up to 160 mT with a two-axis tumbler contained in a three-layer mu-metal shield. Thermal demagnetization (ThD) was carried out progressively in more than 12 steps up to 700°C. Specimens were heated in air using a noninductively wound electric furnace. The stray field during the cooling cycle was reduced to less than 10 nT using a three-layer mu-metal envelope.

Using the vector-demagnetization diagrams for progressive ThD of pilot specimens, the following sites were excluded from the routine demagnetization procedure: (1) progressive change of the vector endpoints did not show a straight trend. Secondary magnetizations were not removed effectively for this case, and (2) stable magnetic direction coincided with the present geomagnetic field direction before tilt correction. Such a component might be of the secondary overprint of the present field.

The routine demagnetization was carried out on sites which survived the above-mentioned criteria. Eight to 12 specimens per each site were demagnetized at three to five steps. The ThD method was used for the sites whose secondary components were erased more effectively by ThD than by AFD. For the sites in which both ThD and AFD were able to detect stable primary components, AFD was adopted because ThD might give rise to breakdown of specimens in consequence of dehydration. In each site, the average paleomagnetic direction of specimens at demagnetization level to produce minimum within-site scatter was adopted as the site-mean direction.

Welded tuffs (NH01-15, HK52) from the Cretaceous Nohi rhyolite and the Tori Formation, the lowermost part of the Neogene System, were generally found to be magnetically stable on ThD. As shown in Figure 3, they showed a single component in remanent magnetization except for unstable components which were erased out in demagnetization at the lower temperature (up to about 200°C). Curie points of many samples expected from the ThD character were about 580°C (see Figure 3a), indicating that the stable remanent magnetization was mainly carried by magnetite with little or no Ti content. In two sites (NH11,13), however, magnetic phase with the highest end of blocking temperature in the range between 580°C and 650°C was identified through ThD (Figure 3b). Since little change was observed in the direction of remanence after demagnetization at a temperature above the Curie point of magnetite (578°C), both magnetite and hematite phases probably express a primary magnetic component [Geissman and Van der Voo, 1980]. In five sites (NH07,08,09,12,15), stable magnetic direction was not recognized on the vector-demagnetization diagram. These sites were excluded from the routine demagnetization.

Sedimentary rocks (tuff, tuffaceous siltstone, and mudstone) were sampled in seven sites from the Neogene System around Yatsuo. It was found that both tuff and tuffaceous siltstone had reliable primary magnetic components through demagnetization tests, in general. Most of the mudstone samples were not acceptable because magnetization of them turned completely unstable after ThD above 300°C. This instability is probably due to secondary remanent magnetization induced by the geomagnetic field in our laboratory. The unstable remanence is thought to be carried by a new magnetic phase produced during ThD. For such samples, AFD was not effective to detect a stable component, either.

Fourteen sites in the Nohi area were found to have a reliable primary magnetic component through the demagnetization test. Since the sources of the error for a paleomagnetic study (tectonic tilt, overprinting of secondary magnetization) have been carefully eliminated through sampling and experimental procedures, each site-mean direction can be regarded as the geomagnetic field direction. Site-mean directions, combined with published data [Itoh, 1986] in the Nohi area, are drawn in Figure 4 and listed in Table 1. The paleomagnetic data have been obtained from three stages: Late Cretaceous, early Miocene (23-15 Ma), and middle Miocene (12-10 Ma).

Ten of 15 sampling sites in Late Cretaceous had stable primary magnetization. Site-mean directions of these sites are distributed in two groups antipodal to each other, and most of them show slight easterly deflection in declinations. There is no time-dependent change in declination or inclination values within the studied section of the Late Cretaceous Nohi rhyolite. The formation-mean direction of the 10 sites is D=17.0°, I=49.8°, k=63.1, and α_{95}=6.1°.

Four of eight sampling sites in the early Miocene stage (23-15 Ma) had stable primary

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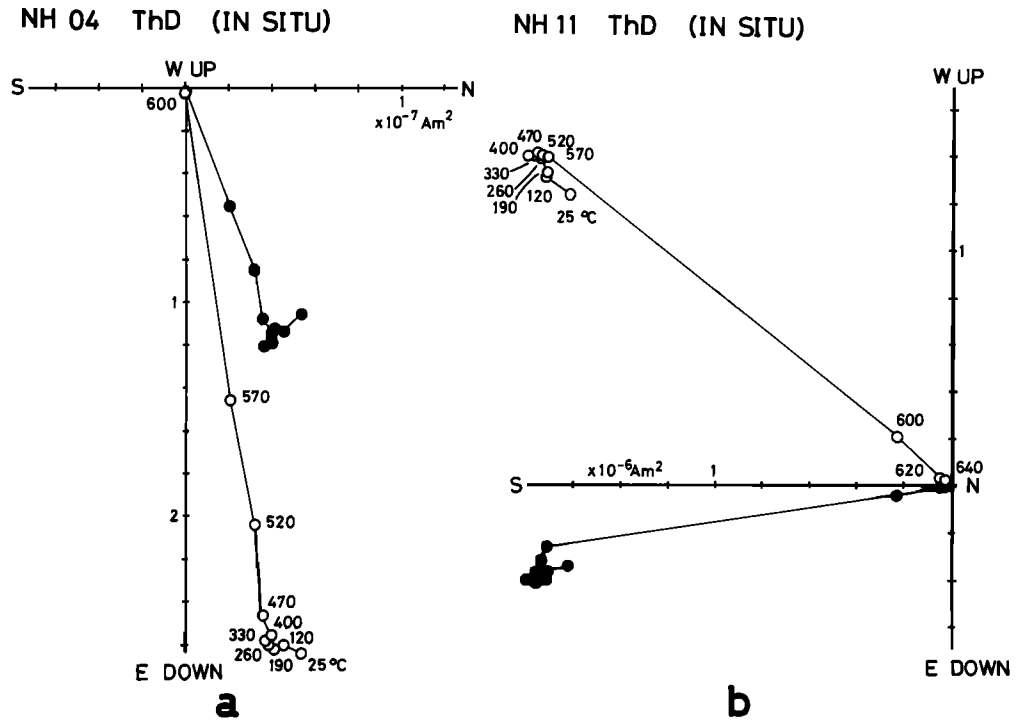


Fig. 3. Typical vector-demagnetization diagrams of progressive thermal demagnetization of welded tuff. Solid (open) circles are projection of vector endpoints on the horizontal (N-S vertical) plane in in-situ coordinates. Unit of coordinates is bulk remanent intensity. Numbers attached to symbols are temperature in degrees Celsius. Progressive change of vector endpoints shows a straight trend indicating stable magnetization. (a) Stable component is erased around the Curie temperature of magnetite (578°C). (b) Magnetic phase with the highest end of blocking temperature in the range between 580°C and 650°C is identified.

magnetization. The site-mean directions of these sites and the four contemporaneous sites of Itoh [1986] are distributed in antipodal directions, and all of them show slight easterly deflections (5°-20°). Inclination values of site-mean directions range from 40° to 75°. The formation-mean direction for eight sites of the early Miocene stage is D=13.0°, I=52.7°, k=59.9, and α_{95}=7.2°.

The data of four sites in the middle Miocene stage (12-10 Ma) were cited from Itoh [1986]. The tilt-corrected site-mean directions coincide almost with the axial geocentric dipole field. As the mean directions in the middle Miocene stage and the axial dipole field direction (D=0°, I=56°) are more than 20° apart before tilt correction (Table 1), the paleomagnetic directions of the four sites are not recent overprint but the primary directions. The formation-mean

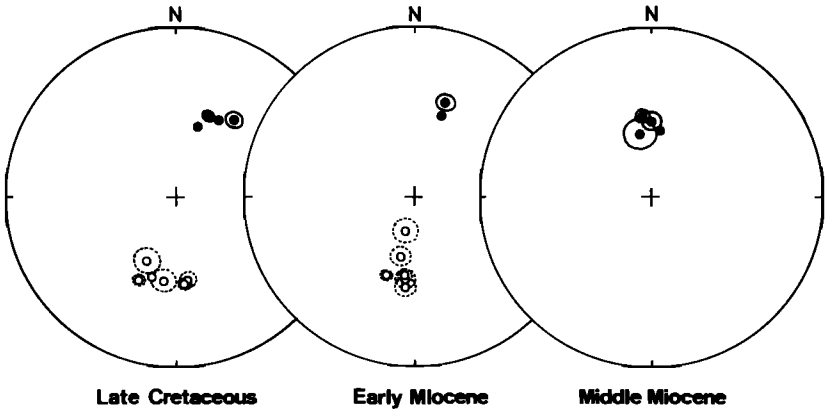


Fig. 4. Equal-area plots of site-mean directions after tilt correction for the Late Cretaceous, early Miocene (23-15 Ma), and middle Miocene (12-10 Ma) stages. Solid (open) symbols represent positive (negative) inclinations. Ovals show 95% confidence circles.

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TABLE 1. Paleomagnetic Site-Mean Directions Obtained From the Nohi Area

Site	Demagnetization Level	D	I	Dc	Ic	N	α_{95}	k	P	PHI	LMD	Lithology
Middle Miocene												
Otogawa Fm. (12.3-9.5 Ma)												
32*	120°C	-38.7	61.2	-6.4	50.8	8	3.2	294.4	N	82.7	5.1	tuff
31*	20mT	-43.4	63.4	-10.4	59.4	11	7.4	39.2	N	81.1	74.2	tuff
16*	10mT	-32.4	75.1	6.9	58.0	10	2.0	572.0	N	84.2	-155.6	tuff
Tenguyama Fm. (12.3 Ma)												
18*	10mT	-40.4	71.8	-0.1	53.4	8	4.6	145.1	N	87.3	-41.1	tuff
Early Miocene												
Higashibessho Fm. (15.5-14.9 Ma)												
05*	20mT	-143.5	-84.9	-172.5	-51.7	12	3.2	190.1	R	82.5	-100.5	tuff
YT04	400°C	-36.9	-63.0	-164.9	-73.1	10	5.9	68.9	R	65.8	156.3	tuff
Kurosedani Fm. (15.5 Ma)												
33*	190°C	171.6	-52.3	-174.0	-45.7	12	4.9	80.1	R	79.3	-72.9	pumice tuff
Iozen Fm. (16.4-15.5 Ma)												
36*	20mT	42.7	58.2	18.3	48.7	8	2.0	790.8	N	73.2	-113.9	welded tuff
Iwaine Fm. (16.4 Ma)												
35*	330°C	70.3	61.4	17.9	41.7	9	4.3	143.1	N	70.2	-98.8	andesite
Nirehara Fm. (23.0-16.4 Ma)												
HK53	400°C	-70.1	-73.6	-166.8	-60.6	9	4.8	114.2	R	78.6	-163.1	tuff
HK50	10mT	-140.6	-72.9	-160.1	-49.8	12	2.7	262.9	R	72.4	-119.0	siltstone
Tori Fm. (23.0 Ma)												
HK52	400°C	160.4	-46.7	-173.5	-49.8	11	4.7	94.2	R	82.0	-87.8	welded tuff
Late Cretaceous												
Nohi rhyolite												
NH01	20mT	-142.7	-46.3	-162.6	-49.3	12	1.9	544.4	R	74.4	-117.2	welded tuff
NH02	550°C	-152.4	-40.6	-155.5	-45.6	12	3.0	212.6	R	67.5	-117.3	welded tuff
NH03	15mT	-156.8	-41.3	-154.9	-56.2	12	6.2	49.3	R	69.8	-143.4	welded tuff
NH04	10mT	68.5	61.8	29.0	47.4	12	1.6	762.2	N	64.5	-123.9	welded tuff
NH05	20mT	62.0	62.1	21.9	47.8	11	2.6	310.6	N	70.2	-117.6	welded tuff
NH06	500°C	69.4	54.3	17.1	54.5	12	1.8	586.4	N	76.1	-133.3	welded tuff
NH10	560°C	174.4	-52.1	172.5	-49.5	12	3.9	124.8	R	81.7	8.6	welded tuff
NH11	20mT	174.7	-47.6	174.7	-47.6	12	3.0	203.6	R	81.6	-8.9	welded tuff
NH13	500°C	-177.2	-61.1	-171.1	-48.8	12	5.7	58.3	R	80.4	-96.3	welded tuff
NH14	400°C	59.8	61.9	36.9	43.2	9	4.1	158.4	N	56.7	-124.5	welded tuff

Numerical ages in the Miocene stages are determined on the basis of age estimations shown in Figure 2. Demagnetization level is in degrees Celsius or millitesla (mT). D, I are mean declination and inclination in degrees before tilt correction; Dc, Ic are mean declination and inclination in degrees after tilt correction; N is number of specimens; α_{95} is radius of 95% confidence circle in degrees; k is Fisher's precision parameter; P is magnetic polarity (N and R indicate normal and reversed polarity, respectively); PHI, LMD are north latitude and east longitude of virtual geomagnetic pole position (north-seeking pole) in degrees.

*Data from Itoh [1986].

direction is D=-2.5°, I=55.6°, k=197.5, and α_{95}=6.6°.

Discussion

The present study has indicated that the formation-means of paleomagnetic directions obtained from the Nohi area show slight easterly deflections in Late Cretaceous and early Miocene. However, the deflection angles are much smaller than those of the previous estimation for southwest Japan. For instance, Otofuji and Matsuda [1987] decided the deflection angle during early Miocene as 60° on the basis of the paleomagnetic data in the San'in area, the central part of southwest Japan. This discrepancy means that a differential rotation had taken place between the eastern part of southwest Japan, including the Nohi area, and the central part of southwest Japan after early Miocene. Figure 5 illustrates the distribution of the Late Cretaceous and early Miocene paleomagnetic data on which progressive demagnetization tests and tilt correction have been performed. As shown in the figure, the possible boundary between the two parts which had differentially rotated is situated in the western neighborhood of the Nohi area.

Paleomagnetically determined boundary is not concordant with the geotectonic division based on the distribution of major fault systems

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[Takeuchi, 1986]. Most pre-Neogene terranes in southwest Japan are composed of complexly folded and faulted accretionary prisms from Jurassic to early Cenozoic. In the boundary zone, however, there has not been found either remarkable faults or fractures to compensate the large amount of differential rotation such as the Tsurugawan-Isewan tectonic line (see Figures 1 and 5). Absence of remarkable faults may indicate that the differential rotation was caused by plastic bending of the terranes composing the Japan arc around the boundary zone.

Figure 6a shows the paleomagnetic declination values during Neogene obtained from the main part (west of the above-mentioned boundary zone) of southwest Japan [Otofuji et al., 1985a], and Figure 6b also shows the Miocene declinations from the Nohi area. As clearly shown in the figure, the declination values of the two parts are quite discordant before 15 Ma, namely, before the initiation of the clockwise rotation of southwest Japan associated with the opening of the Japan Sea. Contrary to this, the paleomagnetic data around 13 to 12 Ma show no significant difference. It is therefore suggested that the differential rotation between the Nohi area and southwest Japan had occurred rapidly during the period between 15 and 12 Ma.

Since the apparent polar wandering path (APWP) during the last 100 m.y. has been proposed from the San'in area in the central part of southwest Japan [Otofuji and Matsuda, 1987], the San'in area is used as a reference to estimate the amount of differential rotation in the eastern part of southwest Japan. The mean paleomagnetic declination (D), inclination (I), and uncertainty in declination (dD) defined by Kellogg and Reynolds [1978] of the central part of southwest Japan are calculated from APWP for the San'in area at the representative location of southwest Japan (35°N, 134°E). For early Miocene, D, I, and dD are 60.0°, 41.8°, and 14.7°, respectively. On the other hand, the mean direction of early Miocene data in the Nohi area is D=12.5° and I=50.5° at the same representative location (dD=11.9°). Using these data, the angle of relative rotation (R) and its uncertainty (dR) defined by Beck [1980] are given as -47.5° and 18.9°, respectively. It is concluded that the Nohi area has undergone a counterclockwise rotation relative to the central part of southwest Japan through 48°. In the same way, the amount of differential rotation between the central part of southwest Japan and the Nohi area since Late Cretaceous is calculated as R=-56.8° (dR=16.6°). The amount of this rotation is comparable to that of the rotation obtained from early Miocene data, suggesting that most of the differential rotation between the Nohi area and southwest Japan took place in the Miocene.

Results of the present study require rapid (less than 3 m.y.) and large amounts of (as much as 48°) differential rotation in southwest Japan during or just after the opening of the Japan Sea. Plausible mechanisms of the differential rotation are presented as the following two models (Figure 7). In the models, rotational movements of the Kanto area in northeast Japan are also taken into account because the zonally arranged pre-Neogene terranes of southwest Japan can be continuously traced through the Nohi area as far as the Kanto area (see Figure 1). The

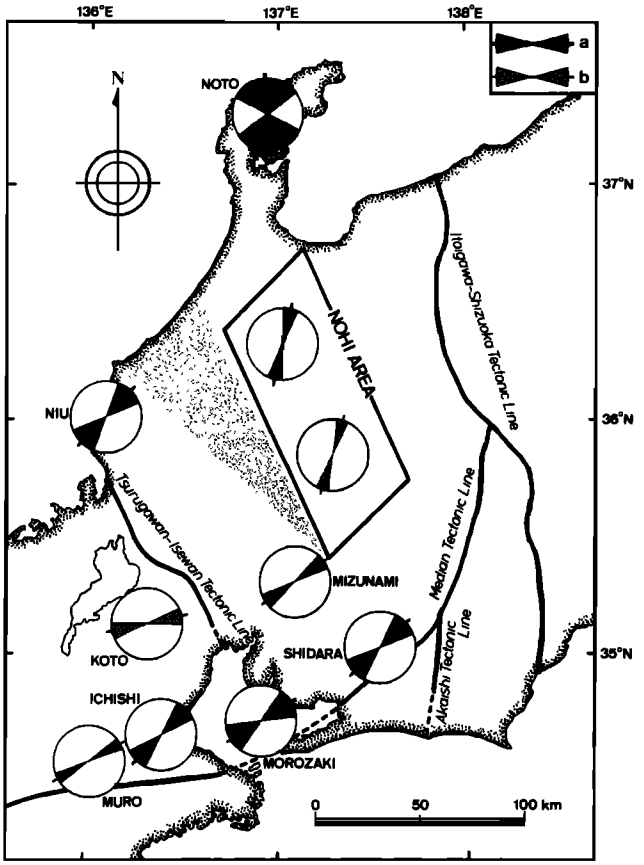


Fig. 5. Possible boundary of the differentially rotated parts in southwest Japan (shaded zone) inferred from the distribution of paleomagnetic directions in (a) early Miocene and (b) Late Cretaceous. Direction of mean declination value is shown as midpoint of the darkened sector in circle for each area. Vertically opposite angles of the darkened sectors indicate uncertainty in declination (dD) defined by Kellogg and Reynolds [1978]. The data in Ichishi, Morozaki, and Mizunami are from Hayashida [1986]; Muro and Shidara are from Otofuji et al. [1985a]; Koto from Fukuma and Torii [1986]; Niu from Nakajima and Hirooka [1986]; and Noto from Hirooka et al. [1986].

continuity indicates that the western part of northeast Japan (west of the Kashiwazaki-Choshi line) and southwest Japan can be regarded as a coherent landmass since, at latest, Jurassic when the main part of the pre-Neogene terranes accreted [Ozawa et al., 1985]. The landmass drifted from the Asian continent on account of the opening of the Japan Sea in the Miocene: (model 1) an intense shear drag force may have been produced on the margin of the rotating arc associated with the fan-shaped opening of the Japan Sea. As southwest Japan was rotated clockwise around a pivot fixed at the western end of the arc, left-lateral shear in the eastern part of the arc caused counterclockwise rotation of the Nohi and the Kanto areas; (model 2) collision of the landmasses on the Philippine Sea plate is a possible cause of the deformation of the island arc. Though the present length of the slab under the Japan arc suggests that the subduction of the

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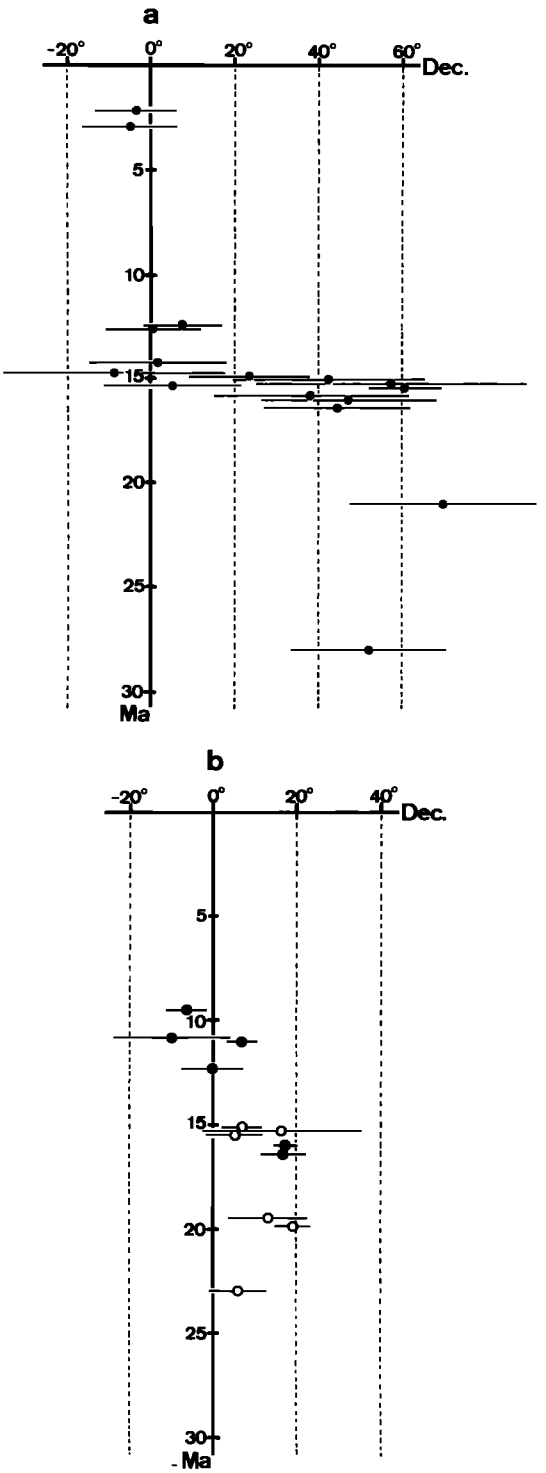


Fig. 6. Declination (at the representative point; 35°N, 134°E) versus age for (a) the main part of southwest Japan and (b) the Nohi area. The error bars are dD [arcsin(sin α_{95}/cos I)]. (a) Mean declination data of rock units [after Otofuji et al., 1985a]. (b) Site-mean declinations. Solid (open) circles show data with normal (reversed) polarity.

Philippine Sea plate can go back to 5 Ma [Matsubara, 1980], the subduction in middle Miocene is surmised from the ages of ophiolite and surrounding rocks in the Mineoka belt [Ogawa, 1983]. Back arc opening of the Japan Sea would be the direct cause of middle Miocene subduction of the Philippine Sea plate. On account of the subduction, the Izu-Bonin arc on the Philippine Sea plate collided against the island arc which had been separated from the Asian continent. Intensely deformed rock bodies in the Fossa Magna, the Cenozoic tectonic zone situated between the Nohi and the Kanto areas, are thought to be the remainders of collided landmasses. Such collision caused counterclockwise rotation of the Nohi area and clockwise rotation of the Kanto area.

The paleomagnetic data obtained from the Kanto area seem to prefer model 2. Hyodo and Niitsuma [1986], who studied paleomagnetism of the early Miocene Series in the Chichibu Basin of the Kanto Mountains, revealed that the mean direction of the sedimentary rocks which were correlated to Blow's zone N8 (16.5-15.5 Ma) showed easterly deflection as much as 94°. They attributed half of the deflection to the rotation of southwest Japan (about 47°), and the remainder to the differential rotation between the Kanto Mountains and southwest Japan after 15 Ma. Therefore it is concluded that the differential rotations of the Nohi area and the Kanto area occurred synchro-

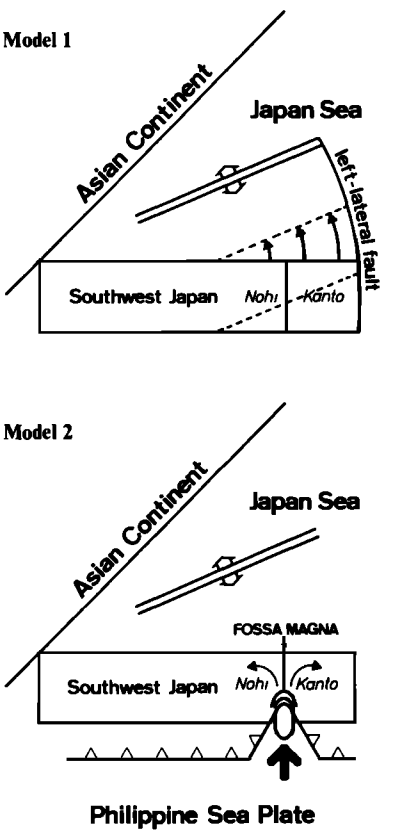


Fig. 7. Two models for the mechanism of the differential rotation as confirmed by paleomagnetic data in the Nohi area. (Model 1) Left-lateral shear is produced on the margin of a rotating arc, resulting in counterclockwise rotation of the Nohi and the Kanto areas. (Model 2) Landmasses on the Philippine Sea plate collide against the Japan arc, resulting in counterclockwise rotation of the Nohi area and clockwise rotation of the Kanto area.

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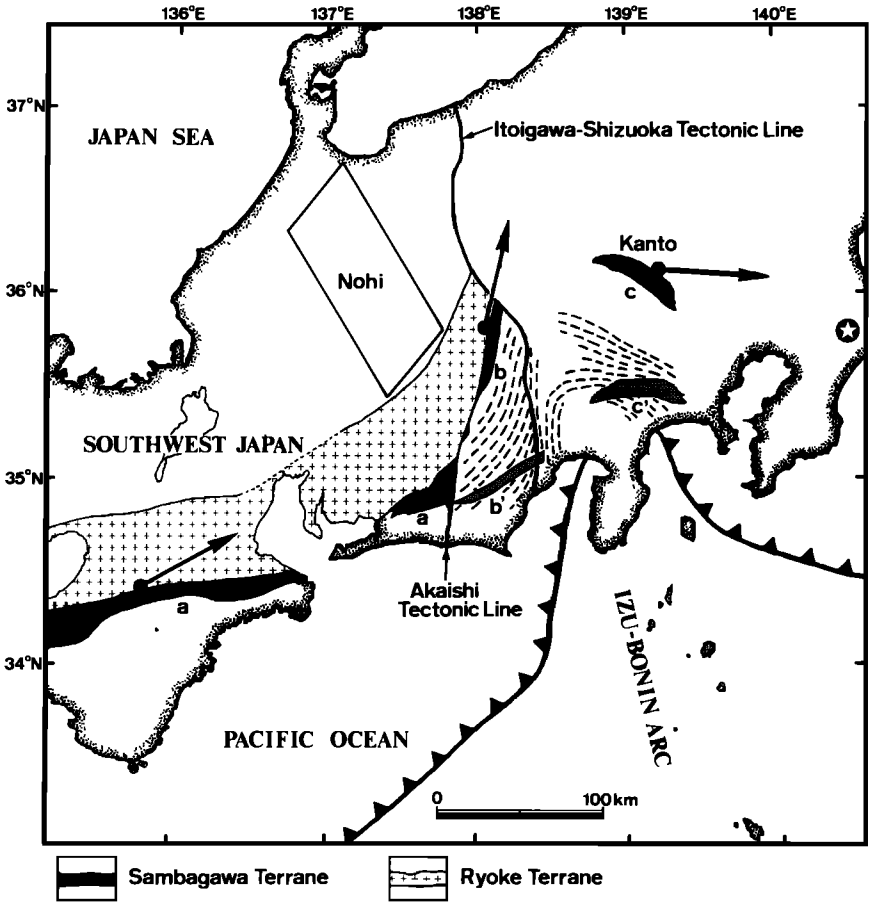


Fig. 8. Rearrangement of the pre-Neogene Sambagawa Terrane before differential rotation of the Nohi and Kanto areas. Segments a, b, and c, which belong to three paleomagnetically distinct parts (central part of southwest Japan, Nohi, and Kanto), show the present arrangement of the Sambagawa Terrane. Structural trends of late Mesozoic and Cenozoic strata around southern Fossa Magna are also shown by short dashes. Arrow attached to each segment shows horizontal component of mean paleomagnetic direction of the part in early Miocene. Data source of the arrow on segment a is Otofuji and Matsuda [1987], on segment b is this study, and on segment c is Hyodo and Niitsuma [1986]. Segments b and c are rotated to the rearranged position of the terrane (segments b' and c') so as to parallel the three arrows on the segments. Star shows the rotation pivot for segment c (pivot C).

nously with the collision of the Izu-Bonin arc in the middle Miocene. Since the collision event occurred, the Nohi and Kanto areas have belonged to different tectonic domains and the eastern limit of the Nohi area has been situated around the Itoigawa-Shizuoka tectonic line (Figure 1).

Based on the paleomagnetic data, a possible paleogeographic reconstruction of the pre-Neogene terranes in southwest Japan and the western part of northeast Japan can be made for the period before the differential rotation. Figure 8 delineates the distribution of the Sambagawa terrane, showing the typical trend of the terranes (Ryoke, Sambagawa, Chichibu, and Shimanto). Northward bending structures around the Fossa Magna are also illustrated by short dashes. On the paleomagnetically distinct three segments of the terrane, horizontal components of the mean paleomagnetic directions in early Miocene are plotted as arrows. Paleomagnetic data of the central part of southwest Japan [Otofuji and Matsuda, 1987], the Nohi area (this study), and the Kanto area [Hyodo and Niitsuma, 1986] are assigned to segments a, b, and c, respectively. The Akaishi tectonic line (ATL) is defined as the boundary of segment a and segment b, since the early Miocene data in the Shidara area, just west of the ATL, show easterly deflection as much as 47° [Otofuji et al., 1985a]. Even if some echelon faults were formed between the rigid Nohi area and segment b of the Sambagawa terrane, the differential rotation associated with them is supposed to have been of a negligible amount. To obtain the paleoposition, segment b is rotated so as to parallel the arrow on it with the arrow on segment a. Then the segment is translated toward segment b' so as to cancel the offset of the terrane along the ATL, which is estimated to be 60 km [Matsushima, 1973]. As for the reconstruction of segment c, it is rotated toward segment c' so as to parallel the arrow on it with the arrow on the segment a around the pivot of

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rotation (pivot C). Pivot C is tentatively situated at 35.8°N, 140.5°E, where the Sambagawa crystalline schist has been confirmed by the results of deep drilling [Yamada et al., 1982]. As clearly shown in Figure 8, reconstructed segments a, b', and c' stand almost on a straight line. This result suggests that the rotational event of the Nohi and the Kanto areas in middle Miocene is equivalent to the formation of the northward bending structure of the pre-Neogene terranes.

During the past decades, many ideas have been proposed for the origin of the northward bending structure around the Fossa Magna. Matsuda [1978] compiled previous works and suggested that the bending structure had been formed during Paleogene time. However, clear evidence to support Paleogene bending was not presented. Hyodo and Niitsuma [1986] attributed the clockwise rotation of the Kanto Mountains, which was indicated by the paleomagnetic data of early Miocene rocks in the Chichibu Basin, to the bending of the pre-Neogene terranes since 6 Ma on account of the collision of the Tanzawa landmass on the Philippine Sea plate. However, paleomagnetic data in the Kanto area provide no information on the initiation or duration of the rotational event after 15 Ma with regard to the pre-Neogene terranes. The present study indicates that the major bending of the pre-Neogene terranes occurred in the period between 15 and 12 Ma associated with the opening of the Japan Sea and collision of the Izu-Bonin arc against the Japan arc.

Conclusions

It was shown that the paleomagnetic declinations in the Nohi area, the eastern part of southwest Japan, had been much smaller than the easterly deflected ones in the central part of southwest Japan from Late Cretaceous to early Miocene. The smaller deflection angle suggests the differential rotation of the eastern part of southwest Japan. The boundary zone of the differential rotation assumed from the available paleomagnetic data is situated in the western neighborhood of the Nohi area. Geologic features seem to suggest that the rotation was caused by plastic bending around the boundary zone. The bending motion, most of which occurred between 15 and 12 Ma, is closely related to the formation of a remarkable northward cuspate structure observed in the pre-Neogene terranes around the eastern part of southwest Japan and the western part of northeast Japan. The possible cause of such an indentation in the Japan arc is the collision of the Izu-Bonin arc on account of surmisable subduction of the Philippine Sea plate in middle Miocene.

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